

# AI-driven risk identification model for infrastructure project: Utilising past project data

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## ABSTRACT

Infrastructure projects are by nature complicated and vulnerable due to unpredictable risks encountered during their execution. The use of expert opinion and qualitative methodologies in traditional risk identification makes them vulnerable to subjectivity and responsiveness, leading to cost overruns, delays, and, ultimately, project failure. Therefore, to improve the accuracy of risk identification, this study utilises historical data in conjunction with AI approaches to develop a data-driven risk identification model. The model determines risk frequency and consequence by matching them to different risk categories in previous projects, considering word semantics. This model also demonstrates the facilitation of proactive decision-making and allows infrastructure project team members to identify risks early and implement mitigation plans. The study also highlights the practical significance of utilising historical data to make risk management strategies for infrastructure projects more reliable and efficient.

## 1. Introduction

Infrastructure projects are pivotal in sculpting communities' economic, social, and environmental fabric. They encompass various physical structures and systems, such as roads, bridges, dams, water supply networks, and energy grids. Infrastructure projects are characterised by factors such as complexity, uncertainty, ambiguity, dynamic interfaces, major governmental or external influences, and long durations (Shamshiri et al., 2023). Evidence suggests that infrastructure projects are typically money pits where funds are 'swallowed up' without delivering adequate returns due to poor risk identification. Infrastructure projects typically cost over a billion US dollars and take years to develop (Ikhsanova et al., 2020). According to Al-Mhdawi et al. (2024) effective infrastructure projects rely heavily on proper risk management. This involves identifying, evaluating, and devising feasible strategies to mitigate adverse consequences (Ariza Flores et al., 2024). According to Carmichael (2023) the successful achievement of project goals within the specified scope, timeline, and budget is closely linked to the effectiveness of these risk management techniques.

Despite the recognized importance of risk management, especially risk identification and assessment, infrastructure projects continue to

face significant challenges in executing effective risk management practices Ford and Abdulla (2023); Forum (2024); Ghorbany et al. (2024). Erfani et al. (2021) are of the view that risk identification and assessment provide the basis for proactive risk management. This proactive approach prevents minor issues from escalating into major problems (Rumane, 2023). For example, during the construction of the Burj Khalifa in Dubai, project managers took meticulous measures to identify and assess critical risks associated with high winds and temperature variations (Omar & Mahdjoubi, 2023). This involved conducting thorough analyses and employing advanced engineering techniques to ensure the structural stability and long-term success of the project. In contrast, the tragic collapse of the I-35 W Mississippi River bridge in Minneapolis in 2007 was linked to previously undetected design flaws, which unfortunately resulted in significant loss of life and widespread societal disruption.

Also project in Crows Nest encountered a structural collapse during the steel fixing and formwork for the capping beam. According to NSW (2021), this incident transpired due to the absence of adequate control measures to mitigate the planning and design risk of unforeseen structural collapse before, during, and after the excavation process. Also, the Barton Highway bridge collapsed during a concrete pour due to poor

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identification and assessment of construction risk during the early stages of the project (NSW, 2021). Over the past few decades, major infrastructure projects in the United States have seen average cost overruns of 28 % (Kwon et al., 2020; Pandithawatta et al., 2024). For instance, the California high-speed rail project has been years behind schedule, and even for the first phase in the Central Valley, expenses have been over \$2.8 billion (Loukaitou-Sideris, 2013). Maryland's purple line provides an additional illustration of a US infrastructure project that exceeded budget and beyond schedule due to poor financial risk evaluation (Finio, 2023). According to Ali et al. (2024), infrastructure projects suffer the most due to poor risk identification.

In addressing these issues, this research introduced a data-driven risk identification methodology using an initial information template for risk built from past data gathered from projects with comparable traits. Using word embedding models, the suggested model accounts for different terminologies in risk data across projects. Through the integration of state-of-the-art data processing techniques with unique model, this study presents a proposed model that seeks to enhance the accuracy, timeliness, and interpretability of risk detection. Thus, it provides a more flexible and scalable answer to the problem of risk management in infrastructure projects, overcoming the shortcomings of earlier methods. Public agencies and project teams can save time by using the knowledge they've gained from similar past initiatives as a risk identification approach. They can use the model to check risk registers for completeness, reliability, and robustness, which is useful for risk auditing.

## 2. Risk identification

Identifying risks is part of the risk management system and is essential to understand various risks and their classifications comprehensively. According to Al-Mhdawi et al. (2024) it is crucial to make well-considered decisions during a project's initial stages. Therefore, thoroughly and accurately identifying the potential outcomes of each risk decision is vital to understanding their impact. According to Chen et al. (2023); Erfani et al. (2023); Ghorbany et al. (2024) risks that go unidentified remain unaddressed and, therefore, pose threats to an infrastructure project's objectives. Effective risk identification helps avoid surprises and encourages the project team to think critically about the project and its associated risks (Rasool et al., 2024; Zou et al., 2008).

Kumar and Mehany (2022) explained that risk identification necessitates a thoughtful and imaginative approach, drawing on the collective experience and expertise of the project team. According to Korytárová and Hromádka (2020) effective risk identification in infrastructure projects relies on the active involvement of key stakeholders, project team members, a designated risk management team, subject matter experts from outside the project, project managers from other projects, and risk management specialists. Akomea-Frimpong et al. (2023); Bria et al. (2024); Rød et al. (2020) explain that the extent of their involvement is shaped by the specific nature and scope of the infrastructure project, and their collective input plays a critical role in ensuring comprehensive risk identification.

However, results are determined by expert opinions and expertise because risk identification sessions take place early in a project when there is little information available about the project (Boamah et al., 2023; Carmichael, 2023; Cavaco & Muniz Junior, 2020). Moreover, Rød et al. (2020) are of the view that risk identification is difficult for public agencies, particularly those with little expertise in infrastructure projects such as roads and highways. It is essential to thoroughly assess the advantages and disadvantages of any specific methods and other alternatives when undertaking infrastructure projects (Boamah et al., 2022; Canesi & Gallo, 2023). Due to the unique nature of each infrastructure project, the method chosen must be tailored to the specific requirements of the situation and determined on a case-by-case basis. Weng et al. (2024) also explained that risk can be identified early and with adequate time to manage the particular risks of each project by using an

automated method that applies historical risk data derived from similar previous projects.

Similar research in the field of risk management is still in its infancy, despite a steady growth in data-driven studies that aim to glean important lessons from the historical data of infrastructure projects (Wu et al., 2022; Zhou et al., 2023). Integrating artificial intelligence (AI) into the identification of risks marks a significant evolution in how infrastructure projects approach risk management (Ivanova et al., 2024). More sophisticated tools are necessary to forecast, evaluate, and mitigate risks in increasingly volatile and complex operational contexts (Du et al., 2023). AI provides a compelling solution to these challenges through its ability to analyze extensive data sets, recognize patterns, and conduct predictive assessments.

According to Okem et al. (2023), infrastructure projects generate a wide range of data daily, including design blueprints, project timelines, procurement records, weather predictions, safety reports, and real-time data from devices. However, traditional techniques for identifying risks from this data are often laborious and prone to human error, leading to delayed or insufficient risk assessments (Nawaz et al., 2020). AI, with its various algorithms, Ivanova et al. (2024) can handle and analyse this data more efficiently and accurately than human analysts. By utilizing historical project data, AI models can be trained to recognise trends and connections that may indicate critical risks in infrastructure projects. Ajayi et al. (2020) explained that AI when integrated into risk identification, has the capability to predict cost overruns by examining variables such as labor productivity, supply chain disruptions, changes in project scope, and resource allocation. Wang et al. (2023); Wu et al. (2022); Wu et al. (2021) also explained that it also analyse historical project schedule data to identify potential causes of delays and recommend strategies to manage schedule risks. Several studies such as Zhou et al. (2023) used deep learning approaches to anticipate delays and risk response in subway infrastructure projects. Peng et al. (2023) also utilized a knowledge graph as a method of operational risk management in tunnel construction. To identify high political risk, Jiang and Martek (2023a, 2023b); Tillmann Sachs and Wanger (2008), used other methods to identify financial risk in infrastructure projects. By presenting a risk data strategy that utilizes natural language processing and word embedding models to create and personalize information registration risk templates for infrastructure projects such as roads and highways, this study adds to the current literature on risk identification.

## 3. Machine learning

### 3.1. Algorithms and techniques

Infrastructure projects produce enormous volumes of unstructured text data, including emails, safety logs, contract documents, reports, and regulatory documents. This data needs to be transformed into structured, machine-readable formats to be used for machine-learning risk identification. To conduct additional analysis and modeling, Machine Learning (ML) approaches convert human language into structured text and subsequently into numerical data (Shirazi & Toosi, 2023; Toğan et al., 2022). Recent research on machine learning approaches has shown that employing specific algorithms that concentrate on semantic meaning and context greatly increases efficiency. In a similar vein, contemporary Deep Learning (DL) and Natural Language Processing (NLP) algorithms efficiently offer ways to transform unstructured text data into machine-readable representations for identification and analysis (Akomea-Frimpong et al., 2023; Khalef & El-Adaway, 2021; Oyedele et al., 2021; Toğan et al., 2022).

The basic operation in text data analysis according to Khalef and El-Adaway (2021) is Tokenization; they further explained that it entails fragmenting text into smaller units like words, phrases, or sentences. This procedure is essential for additional analysis since it clarifies the text's structure (Allah Bukhsh et al., 2020). According to Oyedele et al. (2021), stopword elimination is another crucial method that eliminates

mundane terms like “the,” “is,” and “and.” These words usually don’t add much to the risk identification; therefore, they can be removed to make it more efficient. These techniques greatly enhance the consistency of textual data.

Allah Bukhsh et al. (2020); Moon et al. (2021) also posit that project names, risk factors, and textual locations are among the important elements that is needed to identify risk and can be utilised by using Named Entity Recognition (NER). They further explained that examples of risk-related words that can be used for this approach are “budget overrun,” “delay,” and “contract breach.” Different studies also utilised Vectorization to depict a vector space for numerical calculations in a series of investigations for machine learning techniques. Ledolter and Vander-Valde (2021) also explained that an improved variant of the count-based approach, frequency-based text vectorization that can utilise keywords to distinguish between different types of text data, is Bag of Words (BOW). This can be defined as:

$$TF - IDF(w, d) = TF(w, d) \times IDF(w) \quad (1)$$

Where  $TF(w, d)$  is the term frequency  
 $IDF(w)$  is also given as

$$IDF(w) = \log \frac{N}{1 + [d : w^d]} \quad (2)$$

Although count-based and frequency-based algorithms offer economical and fast ways to vectorize text, they overlook specific crucial elements when it comes to semantic meaning and word sequence (Shamshirgaran et al., 2022). Word embedding models, also known as neural networks, have recently emerged due to developments in ML algorithms. Ayat et al. (2024); Morshed et al. (2024) explained further that the models display every word in the corpus as they capture the underlying semantic meaning.

### 3.2. Application in infrastructure projects for risk identification

In infrastructure projects, risk infrastructure projects frequently entail making judgments in the face of uncertainty. It is imperative for project managers to assess the probability of diverse risks and their possible influence on the project (Nabawy & Gouda Mohamed, 2024). This procedure has historically placed a great deal of reliance on expert opinion and qualitative evaluations, which can be arbitrary and inconsistent (Boamah et al., 2025; Wadley et al., 2021). With the integration of AI, ML algorithms provide data-driven insights that change risk identification decision-making. Infrastructure project contracts, design specifications, risk registries, work and change orders, papers pertaining to injuries and safety, claims and litigation, and a great deal of data pertaining to construction are stored in text formats (Erfani & Cui, 2022; Kim et al., 2022). Using ML models, information processes and document management were automated and enhanced, allowing for the retrieval, and analysis of different types of project documents (Lee et al., 2024). Different studies, such as Oyedele et al. (2021), used an integrated model to retrieve historical data items from power infrastructure projects and categorize their relations according to design and technical specifications. Zhong et al. (2022) also automatically extracted information from quality inspection regulations in infrastructure projects. Checking for regulatory compliance (Zhong et al., 2022), managing constraints (Sonali Pandithawatta et al., 2024), and conducting construction specification reviews using ML approaches all demonstrated promising results that might significantly reduce errors and deficiencies in formulating requirements.

Furthermore, the goal of the research was to create a system that could automatically identify and assess the sentiments and preferences of the users. To gauge the level of support and enthusiasm for infrastructure projects among the general population, this field utilises topic modelling and sentiment analysis applied to text sources. Results show that ML approaches have much promise for use in infrastructure project

risk identification (Livchits et al., 2021; Ahmed Moussa et al., 2024a). According to research, they make it easier and faster to identify and analyse significant texts with a lot of context text.

ML applications also automated the nature of communication between project team members, enabled automatic detection of demand clauses, and identified the responsibilities of each party (Agnisarman et al., 2019; Hallaji et al., 2022; Li et al., 2022). According to Pandithawatta et al. (2024); Peng et al. (2023) by filtering through hundreds of risk records, an ML algorithm can identify a place with an abnormally high number of near-miss risks, therefore enabling the project team to examine and take precautions. Also, the ML model identifies and predicts that new projects would likely go over budget because of problems in the supply chain. Thus, it’s important to take remedial measures as soon as possible (Tirmizi & Arif, 2022; Toğan et al., 2022). By utilizing ML algorithms, the allocation of resources is optimised. In response to unforeseen risks, the model adapts the allocation of resources in real-time.

The literature as a whole suggests that ML models could be useful for learning from past data. Using word embedding algorithms, this research determines the semantic risk similarities between new and existing infrastructure projects, matching risk words from previous projects to identify possible risk items.

## 4. Model development

The approach to developing the risk identification model for infrastructure projects is detailed in this section. The user begins by choosing criteria to narrow down the database of retrieved project data. Risk factors, types, delivery methods, and location are some of the attributes that the user could provide. There is always an option to select all attributes instead of just one. One thing to think about is that if all filters are added, it can drastically reduce the number of projects and identified risks that are retrieved, which could lead to bias in the performance of the risk template. After the risk registers are chosen, the model uses word embedding models to create the risk template. Sorting the risk template by frequency or repercussions as defined by the user. Fig. 1 demonstrates the procedures for the proposed model.

### 4.1. Step 1: customized modelling

According to the user’s preferences, the model creates a register template based on the project risk criteria. Risk factors, types, project delivery methods, and location are defined in the first stage using the provided list. Infrastructure projects such as new roadways, bridges and tunnels, highway reconstruction, and interchanges can all be on the list of project types. Risk factors must also be listed in the templates.

Project sizes can fall within the following categories: below \$500 million, between \$500 million and \$1 billion, and beyond \$1 billion. Common approaches to infrastructure project delivery methods must include design-bid-build, design-build, and public-private partnerships. The initial risk template is generated by retrieving similar projects from the database with the required attributes defined by the infrastructure project characteristics. Because of the tools supplied, users can create risk registers in a variety of ways. Users are able to personalize their risk register by evaluating several risk templates in this way. Table 1 also describes the project information template used from the model.

### 4.2. Second step: similarity matching of risk

The initial stage is to use natural language processing (NLP) modelling to find comparable risks with different terminology since various project teams sometimes use other languages and vocabulary in expressing the risks. Research in natural language processing transforms written language into a numeric format.

Similarity matching of risk compares the two sets of information using established criteria, trends, or past data to determine the degree to

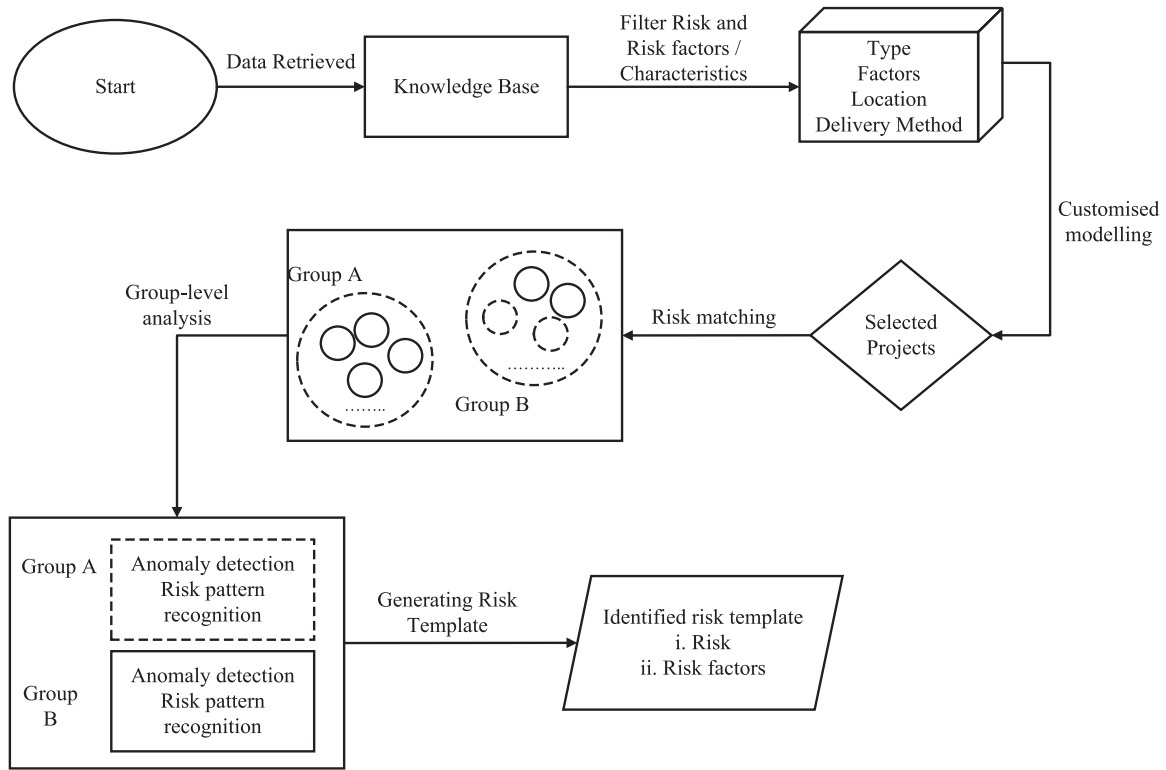


Fig. 1. Risk Identification Model.

Table 1

Project information template.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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| Project Information registration template                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Input:</b> Project size, Risk category, method of delivery, number of risk items, criteria for identification<br><b>Project data:</b> e.g., Rehabilitation of M5 Highway, New South Wales (NSW) Sydney.<br><b>Customised modelling</b><br>i. Size of project: (Choose all / <\$100 M, \$100 M – \$500 M, \$500 m – \$1b, etc)<br>ii. Risk categories: (Choose all / Financial, legal and regulatory, Technical, etc)<br>iii. Delivery: (Choose all / DB, PPP, etc)<br>iv. Number of risk items (Choose 20, 40, 60, etc)<br>v. Criteria for sorting (Cost impact, schedule impact, etc)<br>vi. Selected projects (Type, size, etc)<br><b>Matching of risk</b><br>vii. Defining the risk matching (projects selected)<br>viii. Risk item in project register.<br>ix. Similarity calculation (risk item A, risk item B)<br>x. If the similarity is greater than the agreed percentage,<br>xi. Put risk A and risk item B in the same group.<br><b>Group Level Analysis</b><br>xii. For risk in the same group:<br>a. Risk representative = frequent rep in the group<br>b. Risk type = risk classified type<br>c. Average probability = average (risk probability)<br>d. Average cost impact = average (risk cost impact) ie, from sorting criteria<br>e. Average schedule impact = (ditto) ie, from sorting criteria<br>xiii. Risk registrar template generation<br>a. Risk group results<br>i. Risk template = sorting group risk with the sorting criteria |

which two risks are comparable or relevant. An overarching equation for risk matching using similarity metrics can be stated as the exact mathematical form is approach dependent. the general similarity matching can be utilised as:

$$S(Ra, Rb) = \frac{\sum_{k=1}^n wk.fk(Ra, Rb)^k}{\sum_{k=1}^n wk} \quad (3)$$

Where:

$S(Ri, Rj)$ : The similarity score between risk  $Ra$  and  $Rb$ .

$N$ : Number of attributes utilised for the matching,

$wk$ : The weight assigned based on the significance,

$fk(Ra, Rb)$ : Similarity function. That uses either cosine similarity, Jaccard similarity, or Euclidean distance. The Table 2 provides an example of similarity matching of risk.

To identify risks in infrastructure projects, the calculation for similarity offers a way in matching risk that are similar using diverse terminology from multiple risk registers. When there is no variation in the

**Table 2**  
Example of similarity matching of risk.

| Risk (a)                      | Risk Similarity | Index | Risk (b)                            |
|-------------------------------|-----------------|-------|-------------------------------------|
| Design and structural changes | 1               |       | Design and structural changes       |
| Relocation of utilities       | 1               |       | Relocation of utilities             |
| Delays in government permits  | 0.21            |       | Permits taking longer than Expected |
| Exchange rate fluctuation     | **              |       | Impact on imported materials        |
| Delays in clients' payment    | **              |       | Delays in clients' payment          |
| Exposure to risk materials    | **              |       | Presence of risk substance          |

similarity index between projects, it can be said to be a perfect match.

Choosing a random subset of potential matches is sufficient to establish the cutoff for meaningful matching. The ratio of correct matches to total samples is the formula for accuracy. For example, 75 % similarity can be set as the threshold to risk match, even though the example in Table 3 indicates otherwise. This can be in line with the usual practice of considering the 70th percentile for risk-based estimation at (FHWA, 2014). Python can be used to calculate similarity automatically. All project documents containing risk items must be utilised. Following the user's preference, the code compares the first risk from the first chosen project against all things from the retrieved projects. Risk 1 is grouped together with all other risks in the database that share at least the approved percentage of their characteristics. Once all items in the chosen projects have found a match, the process continues in the second loop, which excludes matches from the first loop. This step generates a list of categorised risks that have been identified differently by separate project teams and have varied language.

#### 4.3. Third step: group-level analysis

The following step is to assess the risks that have already been categorized in order to identify the group's representative text and to determine the average likelihood of occurrence and the repercussions according to the impact of the sorting criteria. It is possible where there are instances where various project teams utilise the same language while describing risk. Each group must be represented by the wording that was most often used by the project teams. The results make it possible to determine the average likelihood and the effect of sorting criteria. Table 4 provides an example of how group-level analysis discovers the final risk item automatically, using an example of identical items in projects with different terminology.

For example, the representative risk chosen is "challenges with construction caused by delayed utility relocation and absence of a clear right of way" since there will be more usage from different project teams. Risk items must have a semantic similarity of 70 % or higher centered on their language to be deemed part of the same group.

In addition, a risk classifier can be useful for determining the optimal risk category for each item, which is necessary for producing organised results. The example categories into which the identified risks can be placed include: "environmental," "structure," "design," "utilities," "highways," "affiliations and stakeholders," "management and grant," "contracting and procurement," and "construction," all must be based on the risk breakdown structure used by selected awarding agency.

First, any unnecessary words or phrases must be eliminated. Secondly, the code encapsulates all potential risks into a vocabulary of

**Table 3**  
Example of Similarity matching threshold selection.

| General similarity matching (100 %) | Sample size | Accuracy (100 %) |
|-------------------------------------|-------------|------------------|
| 100                                 | 60          | 100              |
| 80–100                              | **          | 100              |
| 70–80                               | **          | 82               |
| 60–70                               | **          | **               |
| **                                  | **          | **               |

**Table 4**  
Example of Group-Level Analysis.

| Risk item                                                                                             | Frequency (out of a given number of projects eg: 25) |
|-------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| challenges with construction caused by delayed utility relocation and absence of a clear right of way | 7                                                    |
| Encounter unanticipated or dent to utilities during construction                                      | 4                                                    |
| Delays in relocation                                                                                  | **                                                   |

words. The similarity between risk names and risk categories must be computed after the same procedure is applied to risk categories. The groups must then be given each risk category based on their texts' similarity. Table 5 shows several instances where the risk type for risk items was identified. Although a single risk item may connect to more than one risk type group and defy easy categorisation, the classification aims to identify the most similar group. The risk item "possible changes to design for column" is an example of a combination of "structure" and "design" factors. For example, if "structure" comes out on top, while "design" comes in second, according to the similarity calculation. The top related groups are given the risk.

#### 4.4. Fourth step: generating risk template

Several possible approaches to template customisation involve clustering risk and calculating the average consequences for each cluster. Timetable consequences, average cost or probability, and the frequency of actual initiatives drive prioritisation and sorting. Hence, two primary questions must be addressed to construct a risk registry. Secondly, how many distinct risk elements should be considered in the template? Lastly, what sorting criteria ought to be implemented? To create a risk template according to the user's preferences, the model might take into account different options. Based on the frequency or implications of risks and user construct templates, it is possible to incorporate the top ten, twenty, or thirty items in the risk template, as is allowed by FHWA rules. The decision is application-specific; the template does not impose any numerical constraints. To keep the project team focused on essential risk concerns, the template length ought to include just the right amount of detail without neglecting any major risk. Since this is the case, the risk register needs only ten to thirty elements.

#### 4.5. Risk identification model testing and validation

Due to its critical role in identifying comparable risks across languages, the word embedding technique will form the backbone of the proposed risk identification model. To conduct the study, choosing the optimal Machine Learning method for text-to-vector conversion is best. To initiate the testing procedure, the project information risk register must be conducted or completed for new projects that are not included in the main database. Examples of the testing initiatives are indicated in Table 6.

The next step must involve matching each testing project's risk items to the risk with the highest general similarity based on the identified risk item from the database. The classification of the matching of risks into three categories: "high," "medium," and "low" similarity will be the next

**Table 5**  
Examples of risk classification.

| Risk item            | Risk categories | Similarity | Selected type. |
|----------------------|-----------------|------------|----------------|
| Potential changes to | Environmental,  | 0.441      | Structure      |
| Column design        | Structure       | 0.432      |                |
|                      | Design,         | 0.120      |                |
|                      | Legal           | 0.333      |                |
| ***                  | ***             | ***        |                |

**Table 6**

Examples of project detailed for model testing.

| Project | State ID              | Type of project                     | Size of project (\$) | Risk items |
|---------|-----------------------|-------------------------------------|----------------------|------------|
| I       | New South Wales (NSW) | Highway                             | > 1B                 | 28         |
| II      | Victoria              | Rehabilitation of Selected roads    | \$500 M – 1B         | 15         |
| III     | South Australia       | Repairs on bridge construction      | < \$500 M            | 10         |
| IV      | **                    | Construction of new drainage system | \$500 M – 1B         | 37         |
| ***     | **                    | ***                                 | ***                  | **         |
| **      | **                    | ***                                 | ***                  | **         |

step. Risk items with a high degree of similarity indicate that both projects share subject matter and practical application. If the levels of similarity are medium, then the risks are conceptually similar but have diverse potential uses; if they are low, then the risks are entirely distinct. Table 7 indicates an example of such.

Considering the following when deciding on the final model is very important. The model must first be intelligent; that is, it must correctly identify the highest comparable risk. secondly, the model's index should appropriately represent the similarity level.

In addition, when more than a significant percentage of the risk items across all testing projects are duplicated in the project information registers of at least one database project, the system can be said to be significant. Also, when a significant percentage of risk with a high degree of similarity even employs the same vocabulary and expressions, then the findings can indicate that using past risk data to inform risk identification and assessments of current infrastructure projects is a viable option. By comparing previous projects, this data-driven method will swiftly provide standard risk templates, allowing infrastructure project teams to focus on project-specific risks.

The validation framework should incorporate a set of comparative experiments to thoroughly assess the suggested risk identification model's performance within the framework of infrastructure projects. These tests are necessary for two reasons: first, to ensure that our model is accurate and applicable to a wide range of scenarios; second, to compare its results to those of more conventional expert-based methods and current AI-driven approaches. The accuracy, precision, recall, F1-score, and processing efficiency (in terms of computation time and data preparation needs) were some of the common measures used to evaluate the model's performance. Further evaluation is required to determine the practical usefulness in terms of interpretability and applicability in real project scenarios, taking into account feedback from domain experts.

#### 4.6. Registration template performance

The registration template performance goal is to assess the template's efficacy by analysing selected projects as indicated in Table 6. The process includes;

**Table 7**

Examples of risk similarities.

| Risk (A)                                                                   | Similarity Level | Risk (B)                                                                   |
|----------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------|
| Proximity to high-traffic areas without adequate traffic control measures. | High             | Proximity to high-traffic areas without adequate traffic control measures. |
| Errors in structural analysis                                              | High             | Unstable portal as a result of Poor analysis                               |
| Delays in permit                                                           | Medium           | Permits taking longer than Expected                                        |
| ***                                                                        | Low              | ***                                                                        |

- Create a matrix with the desired number of risks; for example, 15 items.
- Determine which testing project risks are most similar to the template.
- Sort the match's quality as high, medium, or low using a manual cross-check.
- Assess the template's performance by calculating its F<sub>1</sub> score, sensitivity, and accuracy.

It is important to note that a number of results must be defined to compute the performance metrics. If a risk is considered true-positive (TP), it means that it is present in both the testing register and the generated template. This means that the two sets of risks are highly or moderately similar. When entries from the matched risks with a low similarity do not show in the template, a false-negative (FN) occurs. The False positive (FP) also includes all the risks from the template, but none have been used to match any risks in the testing register. Standard measures for evaluating model performance include accuracy, precision, recall, F1-score, and processing efficiency as measured by computation time and data preparation needs. Based on input from domain experts, it is also necessary to assess the practical usefulness in terms of interpretability and application in real-world project circumstances. The template's performance metrics can be equated as;

$$\text{Accuracy} = \frac{\text{True positives (TP)}}{\text{True positives (TP)} + \text{False positives (FP)}} \quad (4)$$

$$\text{Sensitivity} = \frac{\text{True positives (TP)}}{\text{True positives (TP)} + \text{False Negatives (FN)}} \quad (5)$$

$$\text{Therefore, F1 score} = 2 \times \frac{\text{Accuracy} \times \text{Sensitivity}}{\text{Accuracy} + \text{Sensitivity}} \quad (6)$$

Table 8 and Table 9 indicate examples of calculated matching of risk outcomes and template performance, respectively.

Note that sensitivity indicates the performance of the template, which is linked to FN. Accuracy also shows the percentage of risks in the template that have been appropriately identified. F<sub>1</sub> score also strikes a balance between sensitivity and accuracy. The computational efficiency of the template will improve, allowing for faster risk identification in infrastructure projects in real-time or near-real-time by reducing the average processing time. Additional validation shows that the model will give more detailed and practical risk insights, especially when it comes to finding new or compounded risk variables that static or rule-based methods miss.

#### 5. Comparison between the study's AI-driven risk identification model for infrastructure projects and the past 5–10 years

Improved accuracy and predictive capacity are two of the most striking features of contemporary AI risk identification systems. Most models built in the last 5–10 years have relied on more conventional machine learning techniques, such as logistic regression, decision trees, and support vector machines (Isah & Kim, 2022; Nabawy & Gouda Mohamed, 2024; Xia et al., 2017). The methodologies in question, while innovative at the time of their inception, often encounter significant challenges related to scalability and generalizability. Typically developed using datasets pertinent to a specific domain, these methods

**Table 8**

Examples of matching of risk outcome.

| Project ID | High | Medium | Low | TP | FN | F  |
|------------|------|--------|-----|----|----|----|
| I          | 2    | 9      | 18  | 10 | 15 | 17 |
| II         | 9    | 5      | 9   | 13 | 12 | 18 |
| III        | **   | **     | **  | ** | ** | ** |
| IV         | **   | **     | **  | ** | ** | ** |

**Table 9**  
Examples of the template performance.

| Project ID | Accuracy | Sensitivity | F <sub>1</sub> score |
|------------|----------|-------------|----------------------|
| I          | 56.2 %   | 52.3 %      | 51.6 %               |
| II         | 85.3 %   | 79.7 %      | 86.6 %               |
| III        | ***      | ***         | ***                  |
| IV         | ***      | ***         | ***                  |

demonstrate proficiency within a confined operational context; however, they tend to falter when confronted with novel or evolving risk scenarios (Burcar Dunovic et al., 2016; Murphy & Nahod, 2017; Xia et al., 2017).

The AI risk identification model proposed in this study utilizes sophisticated modules trained on extensive and diverse datasets. By analyzing vast amounts of both structured and unstructured data, including real-time updated streams, these models are adept at identifying new risks with remarkable accuracy and predictive power. This capability enables the infrastructure project team to more effectively recognize both overt and covert threats, including those that may have eluded identification in earlier versions of the system.

Another area where there has been tremendous growth is adaptation, according to (Moussa et al., 2024b). The capacity to automatically adapt was frequently missing from studies conducted in the last ten years (Moussa et al., 2024b; Wisetjindawat et al., 2017; Yazdani-Chamzini, 2014; Zhu et al., 2021). When new data or risk variables surfaced, these already trained models needed a lot of work to be retrained or adjusted. Nevertheless, adaptive learning techniques are incorporated into the study's AI risk identification model. The proposed model will be able to automatically and continuously update their understanding through techniques such as reinforcement learning. The capacity to adjust in real-time gives it an advantage over previous models in dynamic environments where they cannot keep up with the pace of change.

A prevalent concern raised about earlier studies was their lack of (Maghsoudi et al., 2015; Mazher et al., 2018; Shi et al., 2015; Sun et al., 2022; Xia et al., 2017). The absence of transparency in the decision-making process led human analysts to question the reliability of the

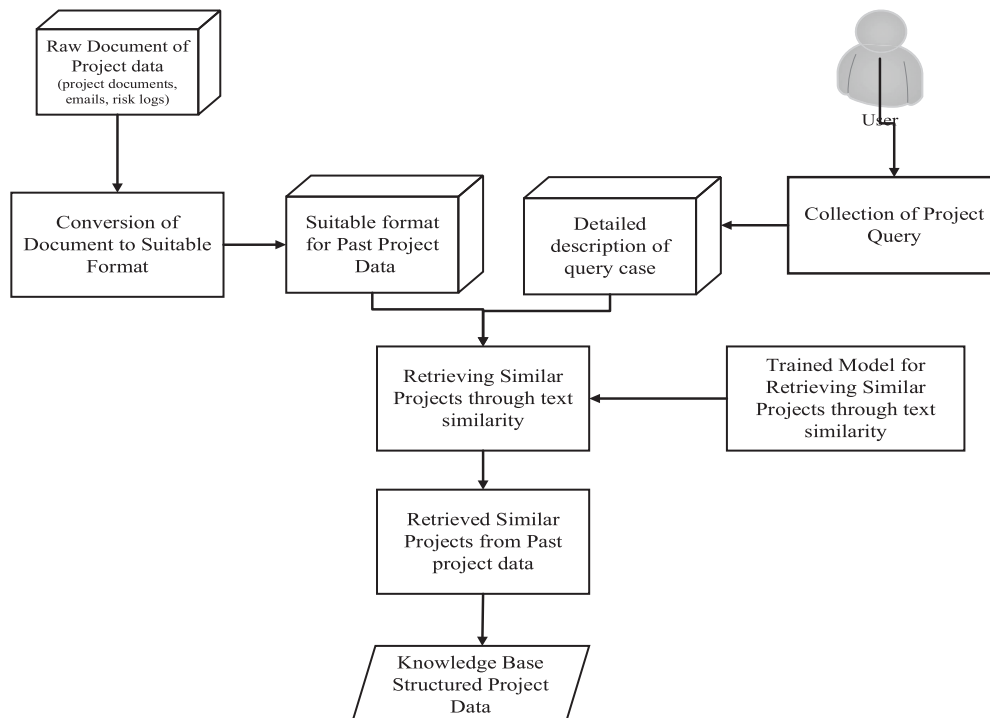
results generated by these models. In response, the study proposes an AI risk identification model that significantly enhances model transparency. The proposed AI risk identification model will now incorporate interpretability frameworks as a fundamental feature. These recent findings contribute to greater transparency and accountability by elucidating the reasoning behind AI-generated risk assessments for decision-makers.

A notable distinction between the AI risk identification model presented in the study and previous methodologies is its ability to manage a diverse range of data. While structured data from specific sources has historically been useful for risk identification, it often proved insufficient to capture the full spectrum of potential risks. In contrast, the AI risk identification model utilised in this study leverages multimodal learning capabilities, enabling the integration and analysis of data from various unstructured data, including text, site meetings, risk logs, project documents, etc. This comprehensive approach enhances their ability to identify complex or cross-domain risks, as indicated in Fig. 2. This study also produced a new data processing pipeline, which enhances the preparation of raw infrastructure data for analysis. This involves state-of-the-art methods for transforming and organising data from various sources, as shown in Fig. 2. Given the wide variation in data quality among infrastructure systems, these processes have added significance.

Fig. 3 also shows that the advanced AI risk identification models have greatly increased in quality over the past few years. Fig. 3 shows a huge improvement over the previous 5–10 years of research and tools, which gave important insights and capabilities. This enhanced model is more than just a tool for risk management; it is a revolutionary technology that is changing how infrastructure project teams perceive, predict, and respond to risk. It is more accurate, adaptable, transparent, and has better data integration and ethical alignment. The model also uses the output of the retrieved structure data as its input.

## 6. Discussion and final implementation

For infrastructure projects, the proposed risk identification model will use previous infrastructure project data to generate the templates. This data-driven risk identification method will have many benefits over



**Fig. 2.** Retrieval from Unstructured Data to Structured Data.

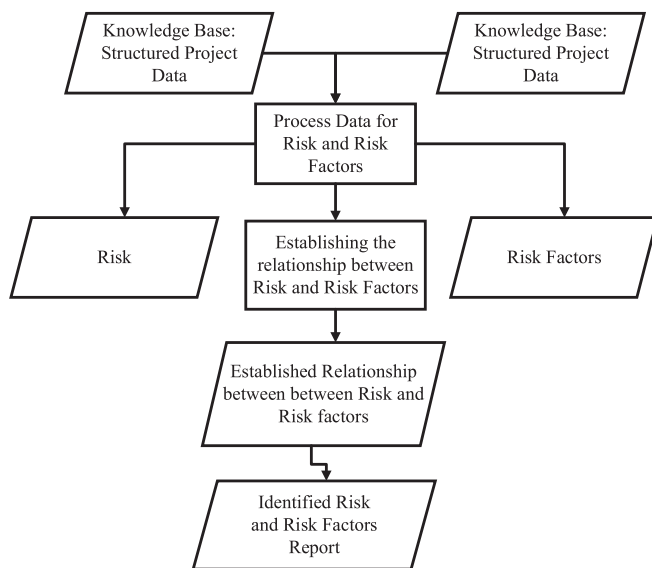


Fig. 3. Redefined process for risk identification model.

traditional methods that rely on expert opinion. The application of AI methodologies underpins the model, enabling the system to analyze vast amounts of unstructured data. This data is sourced from a variety of documents related to projects, contracts, and risks. To standardize and retrieve meaningful information from the text, preprocessing techniques such as named entity recognition must be employed.

Firstly, this model delivers a significant way to change the template according to your user's project needs. In addition to the occurrence of risk items in historical data, it also allows the user to assess the effects of the sorting criteria. In comparison to more conventional methods of risk identification, the proposed model is more efficient because the findings of risk identification from traditional methods are heavily impacted by expert prejudice. Nonetheless, by utilizing empirical data, the data-driven strategy will speed up the processing while capturing the subjectivity and ambiguity of experts' judgments. The risk landscape was structured using a multi-tiered taxonomy that differentiates between various types of hazards, including risks and risk factors. This taxonomy provides a systematic perspective to assess both existing and emerging risks.

The Risk Scoring System, which considers both the likelihood of a risk occurring and its potential impact, is a crucial component of the implementation process. The infrastructure project team can utilize this rating system to prioritize mitigation activities based on quantifiable data rather than intuition. Each risk is assigned a dynamic score that can be adjusted in real-time as additional information becomes available.

The proposed model simplifies incorporating comparable past projects' average outcomes and risks. The model's adaptability will be used for different kinds of projects involving tabular databases. While this research makes some theoretical contributions, the proposed model will be very effective for actual roads, bridges, and tunnel projects.

When implementing the model, it is important to consider the possibility of bias in low-data-point conditions. However, regarding accuracy and sample size, the model is accurate. As a first stage in performing risk assessments, it serves as an advisory tool for infrastructure project teams. The absence of historical post-risk data is one of the key limitations of the present investigation. Considering the sorting criteria's actual values, in addition to the project's success, will greatly enhance the current study's worth.

The field of risk identification in infrastructure projects has recently witnessed a significant influx of artificial intelligence (AI) technology. Although this shows that AI methods are getting better, our research stands out because we included new improvements to model

architecture and data processing. To better identify complicated, high-dimensional risk indicators, we provide a hybrid technique that combines machine learning with predictive algorithms. The model outperforms its predecessors in risk similarity and group level analysis, as confirmed by thorough benchmarking on several real-world datasets. Our approach stands out from current mainstream methodologies because to these changes, which together create a risk detection framework that is more robust and easier to understand.

## 7. Conclusions, limitations, and future studies

This research provided an alternative to depending only on expert opinion by providing a risk identification modeling that will utilize historical infrastructure project data. This data-driven model may examine risk registries of comparable past infrastructure projects for risk using a variety of terminology. Critical risks can be identified by the model by taking into account project specifics, in addition to their prevalence and other sorting criteria effects. The model's applicability can be validated by analysing the template across different examples of infrastructure projects. The model helps project teams automatically identify a higher percentage of the usual risk items. Given the significant similarity between the risk items for infrastructure projects, the proposed model could serve as a starting point for risk studies, allowing project teams to gain insight from comparable historical projects. Every infrastructure project is different; therefore, the template can be fine-tuned and finalised based on the team's best criteria. Reviewing infrastructure projects post risk data that incorporates the actual effect of what transpired and the response used in new projects will help identify future studies that are necessary to address the limitations of the proposed solution. The examples provided in this study are based on assertions; therefore, it is recommended that future studies utilize actual infrastructure project (*Roads, highways, tunnels, etc.*) data in training and testing the model and enhance the registers' volume for comparison and recommended to perform a sensitivity analysis on how the model's performance varies across situations.

## CRediT authorship contribution statement

**Fredrick Ahenkora Boamah:** Conceptualization, Methodology, Validation, Investigation, Writing – original draft, Writing – review & editing. **Xiaohua Jin:** Writing – review & editing, Supervision. **Sepani Senaratne:** Writing – review & editing, Supervision. **Srinath Perera:** Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Data availability

Data will be made available on request.

## References

- Agnisarman, S., Lopes, S., Chalil Madathil, K., Piratla, K., & Gramopadhye, A. (2019). A survey of automation-enabled human-in-the-loop systems for infrastructure visual inspection [Review]. *Automation in Construction*, 97, 52–76. <https://doi.org/10.1016/j.autcon.2018.10.019>

- Ajayi, A., Oyedele, L., Owolabi, H., Akinade, O., Bilal, M., Davila Delgado, J. M., & Akanbi, L. (2020). Deep learning models for health and safety risk prediction in power infrastructure projects. *Risk Analysis*, 40(10), 2019–2039.
- Akomea-Frimpong, I., Dzagli, J. R. A. D., Eluerkeh, K., Bonsu, F. B., Opoku-Brafi, S., Gyimah, S., Asuming, N. A. S., Atibila, D. W., & Kukah, A. S. (2023). A systematic review of artificial intelligence in managing climate risks of PPP infrastructure projects [Review]. *Engineering, Construction and Architectural Management*. <https://doi.org/10.1108/ECAM-01-2023-0016>
- Al-Mhdawi, M., O'Connor, A., Qazi, A., Rahmian, F., & Dacre, N. (2024). Review of studies on risk factors in critical infrastructure projects from 2011 to 2023. *Smart and Sustainable Built Environment*.
- Ali, Y., Sabir, M., & Hassan, M. S. (2024). Risk management in railways construction investments: A case study of china-pakistan economic corridor (CPEC) ML-1 project. *Transport and Telecommunication Journal*, 25(2), 187–199. <https://doi.org/10.2478/ttj-2024-0014>
- Allah Bukhsh, Z., Stipanovic, I., Saeed, A., & Doree, A. G. (2020). Maintenance intervention predictions using entity-embedding neural networks [Article]. *Automation in Construction*, 116, Article 103202. <https://doi.org/10.1016/j.autcon.2020.103202>
- Ariza Flores, V. A., de Sousa, F. O., & Oda, S. (2024). Enhancing Risk Management in Road Infrastructure Facing Flash Floods through Epistemological Approaches [Article]. *Buildings*, 14(7), Article 1931. <https://doi.org/10.3390/buildings14071931>
- Ayat, M., Ullah, M., Pervez, Z., Lawrence, J., Kang, C. W., & Ullah, A. (2024). Analyzing the differential impact of variables on the success of solicited and unsolicited private participation in infrastructure projects using machine learning techniques. *Engineering, Construction, and Architectural Management*. <https://doi.org/10.1108/ECAM-01-2024-0134>
- Boamah, F. A., Jin, X., Senaratne, S., & Perera, S. (2025). Transition from traditional knowledge retrieval into AI-powered knowledge retrieval in infrastructure projects: A literature review. *Infrastructures*, 10(2). <https://doi.org/10.3390/infrastructures10020035>
- Boamah, F. A., Zhang, J., Cao, Z., & Horbanenko, O. (2022). An empirical study on the sharing of tacit knowledge by construction project workers in sub-Saharan Africa. *Knowledge Management Research & Practice*, 1–13. <https://doi.org/10.1080/14778238.2022.2105757>
- Boamah, F. A., Zhang, J., Danso, N. A. A., Alam, S., Naqvi, K., & Tariq, M. (2023). Predicting the reward system of knowledge sharing in the industrialized engineering sector based on regulatory mechanisms. *Iranian Journal of Science and Technology, Transactions of Civil Engineering*, 47(5), 3199–3210. <https://doi.org/10.1007/s40996-023-01093-y>
- Bria, T. A., Chen, W. T., Muhammad, M., & Rantelembang, M. B. (2024). Analysis of Fatal Construction Accidents in Indonesia—A Case Study [Article]. *Buildings*, 14(4), Article 1010. <https://doi.org/10.3390/buildings14041010>
- Burcar Dunovic, I., Radujkovic, M., & Vukomanovic, M. (2016). Internal and external risk based assessment and evaluation for the large infrastructure projects [Article]. *Journal of Civil Engineering and Management*, 22(5), 673–682. <https://doi.org/10.3846/13923730.2015.1128479>
- Canesi, R., & Gallo, B. (2023). Risk Assessment in Sustainable Infrastructure Development Projects: A Tool for Mitigating Cost Overruns. *Land*, 13(1). <https://doi.org/10.3390/land13010041>
- Carmichael, D. G. (2023). *Risk and systems : With applications in infrastructure project management*. CRC Press.
- Cavaco, E., & Muniz Junior, J. (2020). Knowledge-based risk management model: Application in energy infrastructure projects. *Brazilian Journal of Operations & Production Management*, 17(4), 1–16. <https://doi.org/10.14488/BJOPM.2020.046>
- Chen, D., Zhou, J., Duan, P., & Zhang, J. (2023). Integrating knowledge management and BIM for safety risk identification of deep foundation pit construction. *Engineering, Construction, and Architectural Management*, 30(8), 3242–3258. <https://doi.org/10.1108/ECAM-10-2021-0934>
- Du, X., Yu, Y., Ahenkora, B. F., & Pang, Y. (2023). Decoupling economic growth from building embodied carbon emissions in China: A nighttime light data-based innovation approach. *Sustainable Production and Consumption*, 43, 34–45. <https://doi.org/10.1016/j.spc.2023.10.011>
- Erfani, A., & Cui, Q. (2022). Predictive risk modeling for major transportation projects using historical data. *Automation in Construction*, 139, Article 104301. <https://doi.org/10.1016/j.autcon.2022.104301>
- Erfani, A., Cui, Q., & Cavanaugh, I. (2021). An empirical analysis of risk similarity among major transportation projects using natural language processing. *Journal of Construction Engineering and Management*, 147(12). [https://doi.org/10.1061/\(asce\)co.1943-7862.0002206](https://doi.org/10.1061/(asce)co.1943-7862.0002206)
- Erfani, A., Ma, Z., Cui, Q., & Baecher, G. B. (2023). Ex post project risk assessment: Method and empirical study [Article]. *Journal of Construction Engineering and Management*, 149(2), Article 04022174. <https://doi.org/10.1061/JCEMD4.COENG-12588>
- FHWA. (2014). Major Projects Financial Plan.
- Finio, N. (2023). Gentrification and Business Closures in Maryland's Purple Line Corridor. *Journal of Planning Education and Research*. <https://doi.org/10.1177/0739456X231187119>
- Ford, M. J., & Abdulla, A. (2023). New methods for evaluating energy infrastructure development risks. *Risk Analysis*, 43(3), 624–640. <https://onlinelibrary.wiley.com/doi/10.1111/risa.13727>
- Forum, W. E. (2024). Report: Mitigating Political and Regulatory Risk for Successful Infrastructure Projects.
- Ghorbany, S., Yousefi, S., & Noorzai, E. (2024). Evaluating and optimizing performance of public-private partnership projects using copula Bayesian network [Article]. *Engineering, Construction and Architectural Management*, 31(1), 290–323. <https://doi.org/10.1108/ECAM-05-2022-0492>
- Hallaji, S. M., Fang, Y., & Winfrey, B. K. (2022). Predictive maintenance of pumps in civil infrastructure: State-of-the-art, challenges and future directions [Review]. *Automation in Construction*, 134, Article 104049. <https://doi.org/10.1016/j.autcon.2021.104049>
- Ikhshanova, L. R., Shigapova, R. R., Koczar, J., Agliullina, Z. I., Agliullina, M. I., & Syslova, M. E. (2020). Infrastructure Projects and Transport System Financing in Russia. Regional Economic Development in Russia: Institutions, Regulations, and Structural Transformations.
- Isah, M. A., & Kim, B. S. (2022). Assessment of risk impact on road project using deep neural network [Article]. *KSCSE Journal of Civil Engineering*, 26(3), 1014–1023. <https://doi.org/10.1007/s12205-021-1312-2>
- Ivanova, M., Grims, M., Karas, D., Höfinger, G., Bornschlegel, M. X., & Hemmje, M. L. (2024). An AI-Based Approach to Identify Financial Risks in Transportation Infrastructure Construction Projects. *IFIP International Conference on Artificial Intelligence Applications and Innovations*.
- Jiang, W., & Martek, I. (2023a). The mediating role of infrastructure vulnerability on the political risk of foreign infrastructure investment in developing countries: Insights from Brazil and China. *Journal of Infrastructure Systems*, 29(3). <https://doi.org/10.1061/jitsee4.iseng-2151>
- Jiang, W., & Martek, I. (2023b). Strategies for managing the political risk of investing in infrastructure projects, in developing countries. *Engineering, Construction and Architectural Management*. <https://doi.org/10.1108/ecam-12-2021-1072>
- Khalef, R., & El-Adaway, I. H. (2021). Automated identification of substantial changes in construction projects of airport improvement program: Machine learning and natural language processing comparative analysis [Article]. *Journal of Management in Engineering*, 37(6), Article 959. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000959](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000959)
- Kim, Y., Bang, S., Sohn, J., & Kim, H. (2022). Question answering method for infrastructure damage information retrieval from textual data using bidirectional encoder representations from transformers. *Automation in Construction*, 134, Article 104061. <https://doi.org/10.1016/j.autcon.2021.104061>
- Korytárová, J., & Hromádka, V. (2020). Risk assessment of large-scale infrastructure projects—Assumptions and context. *Applied Sciences*, 11(1). <https://doi.org/10.3390/app11010109>
- Kumar, S., & Mehany, M. S. H. M. (2022). Developing a sustainability and resilience monitoring scheme for infrastructure projects using sustainable development goals (SDGs). *Construction Research Congress 2022*.
- Kwon, N., Song, K., Ahn, Y., Park, M., & Jang, Y. (2020). Maintenance cost prediction for aging residential buildings based on case-based reasoning and genetic algorithm [Article]. *Journal of Building Engineering*, 28, Article 101006. <https://doi.org/10.1016/j.jobe.2019.101006>
- Ledolter, J., & VanderVelde, L. S. (2021). Word distributions: Document-term matrices of word frequencies and the “bag of words” representation. In (pp. 49–61). SAGE Publications, Incorporated. <https://doi.org/10.4135/9781071878828.n8>
- Lee, J., Ahn, S., Kim, D., & Kim, D. (2024). Performance comparison of retrieval-augmented generation and fine-tuned large language models for construction safety management knowledge retrieval. *Automation in Construction*, 168, Article 105846. <https://doi.org/10.1016/j.autcon.2024.105846>
- Li, C., Zhou, J., & Liu, F. (2022). Analysis of risk assessment of overseas infrastructure projects integrating BP-ANN algorithm. *Security and Communication Networks*, 2022, 1–13. <https://doi.org/10.1155/2022/7864665>
- Livchits, V., Mironova, I., Tischenko, T., Frolova, M., & Shvetsov, A. (2021). Risk accounting in calculations of public efficiency of large-scale infrastructure projects. 2021 14th International Conference Management of large-scale system development (MLSD).
- Loukaitou-Sideris, A. (2013). New rail hubs along high-speed rail corridor in California: Urban design challenges. *Transportation Research Record*, 2350(1), 1–8. <https://doi.org/10.3141/2350-01>
- Maghsoudi, S., Duffield, C., & Wilson, D. (2015). Innovation evaluation: Past and current models and a framework for infrastructure projects [Article]. *International Journal of Innovation Science*, 7(4), 281–297. <https://doi.org/10.1108/IJIS-07-04-2015-B005>
- Mazher, K. M., Chan, A. P. C., Zahoor, H., Khan, M. I., & Ameyaw, E. E. (2018). Fuzzy integral-based risk-assessment approach for public-private partnership infrastructure projects [Article]. *Journal of Construction Engineering and Management*, 144(12), Article 04018111. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001573](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001573)
- Moon, S., Lee, G., Chi, S., & Oh, H. (2021). Automated construction specification review with named entity recognition using natural language processing. *Journal of Construction Engineering and Management*, 147(1). [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001953](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001953)
- Morshed, M. F., Fattah, M. A., Morshed, S. R., & Sydnunaher, S. (2024). Assessment and prediction of mega-infrastructure projects on rural ecosystems using machine learning algorithms. *Environment, Development and Sustainability*. <https://doi.org/10.1007/s10668-024-05133-z>
- Moussa, A., Ezzeldin, M., & El-Dakhkhni, W. (2024a). Predicting and managing risk interactions and systemic risks in infrastructure projects using machine learning. *Automation in Construction*, 168, Article 105836. <https://doi.org/10.1016/j.autcon.2024.105836>
- Moussa, A., Ezzeldin, M., & El-Dakhkhni, W. (2024b). Predicting and managing risk interactions and systemic risks in infrastructure projects using machine learning [Article]. *Automation in Construction*, 168, Article 105836. <https://doi.org/10.1016/j.autcon.2024.105836>
- Murphy, M. E., & Nahod, M. M. (2017). Stakeholder competency in evaluating the environmental impacts of infrastructure projects using BIM [Article]. *Engineering,*

- Construction and Architectural Management, 24(5), 718–735. <https://doi.org/10.1108/ECAM-07-2015-0106>
- Nabawy, M., & Gouda Mohamed, A. (2024). Risks assessment in the construction of infrastructure projects using artificial neural networks. *International Journal of Construction Management*, 24(4), 361–373.
- Nawaz, A., Su, X., Din, Q. M. U., Khalid, M. I., Bilal, M., & Shah, S. A. R. (2020). Identification of the h&s (Health and safety factors) involved in infrastructure projects in developing countries-a sequential mixed method approach of OLMT-project. *International journal of environmental research and public health*, 17(2), 635. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7014267/pdf/ijerph-17-00635.pdf>.
- NSW, S. (2021). Partial building collapse.
- Okem, E. S., Ukpouju, E. A., David, A. B., & Olurin, J. O. (2023). Advancing infrastructure in developing nations: A synthesis of AI integration strategies for smart pavement engineering. *Engineering Science & Technology Journal*, 4(6), 533–554.
- Omar, H., & Mahdjoubi, L. (2023). Practical solutions for improving the suboptimal performance of construction projects using Dubai construction projects as an example [Article]. *Engineering, Construction and Architectural Management*, 30(6), 2185–2205. <https://doi.org/10.1108/ECAM-10-2021-0956>
- Oyedele, A., Ajayi, A., Oyedele, L. O., Delgado, J. M. D., Akanbi, L., Akinade, O., Owolabi, H., & Bilal, M. (2021). Deep learning and Boosted trees for injuries prediction in power infrastructure projects. *Applied Soft Computing*, 110, Article 107587. <https://doi.org/10.1016/j.asoc.2021.107587>
- Pandithawatta, S., Ahn, S., Rameezdeen, R., Chow, C. W. K., & Gorjian, N. (2024). Systematic literature review on knowledge-driven approaches for construction safety analysis and accident prevention [Review]. *Buildings*, 14(11), Article 3403. <https://doi.org/10.3390/buildings14113403>
- Peng, F. L., Qiao, Y. K., & Yang, C. (2023). Building a knowledge graph for operational hazard management of utility tunnels [Article]. *Expert Systems with Applications*, 223, Article 119901. <https://doi.org/10.1016/j.eswa.2023.119901>
- Rasool, I. I. A., Mohammed, H. A., Ahmed, M. N., Nile, B. K., Shaban, A. M., & Alqarawee, L. S. R. (2024). Building a system for risk management in infrastructure projects. Melville.
- Rød, B., Lange, D., Theodoridou, M., & Pursiainen, C. (2020). From risk management to resilience management in critical infrastructure [Article]. *Journal of Management in Engineering*, 36(4), Article 04020039. [https://doi.org/10.1061/\(ASCE\)ME.1943-5479.0000795](https://doi.org/10.1061/(ASCE)ME.1943-5479.0000795)
- Rumane, A. R. (2023). *Risk management applications to sustain quality in projects : A practical guide*. CRC Press.
- Shamshirgaran, A., Hosseini Nourzad, S. H., Keshkar, H., Golabchi, M., & Sadeghi, M. (2022). Large-scale automated sustainability assessment of infrastructure projects using machine learning algorithms with multisource remote sensing data. *Journal of Infrastructure Systems*, 28(4). [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000703](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000703)
- Shamshiri, A., Ryu, K. R., Park, J. Y., & Choi, K. (2023). Project phase-based contingency adjustment framework for infrastructure renewal projects. *Journal of Construction Engineering and Management*, 149(2). <https://doi.org/10.1061/jcemd4.Coeng-12311>
- Shi, Q., Liu, Y., Zuo, J., Pan, N., & Ma, G. (2015). On the management of social risks of hydraulic infrastructure projects in China: A case study [Article]. *International Journal of Project Management*, 33(3), 483–496. <https://doi.org/10.1016/j.ijproman.2014.06.003>
- Shirazi, D. H., & Toosi, H. (2023). Deep multilayer perceptron neural network for the prediction of Iranian Dam project delay risks. *Journal of Construction Engineering and Management*, 149(4). <https://doi.org/10.1061/JCEMD4.COENG-12367>
- Sun, G., Sun, J., & Li, F. (2022). Influencing factors of early termination for PPP projects based on multicase grounded theory [Article]. *Journal of Construction Engineering and Management*, 148(11), Article 04022120. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002388](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002388)
- Tillmann Sachs, R. T., and Daniel Wanger. (2008). The Quantification and Financial Impact of Political Risk Perceptions On Infrastructure Projects in Asia.
- Tirmizi, S. A. A., & Arif, F. (2022). Conceptual Approach for the Use of Artificial Intelligence for Contractual Risk Assessment in Infrastructure Projects † [Article]. *Engineering Proceedings*, 22(1), Article 12. <https://doi.org/10.3390/engproc202202012>.
- Toğan, V., Mostofi, F., Ayözen, Y. E., & Behzat Tokdemir, O. (2022). Customized AutoML: An Automated Machine Learning System for Predicting Severity of Construction Accidents [Article]. *Buildings*, 12(11), Article 1933. <https://doi.org/10.3390/buildings12111933>.
- Wadley, D. A., Han, J. H., & Elliott, P. G. (2021). Infrastructure planning in Queensland, Australia: Risk appraisal of high voltage overhead transmission lines by property developers and homeowners. *Planning Practice & Research*, 36(1), 41–58.
- Wang, M., Hu, D., Chen, J., & Li, S. (2023). Underground infrastructure detection and localization using deep learning enabled radargram inversion and vision based mapping. *Automation in Construction*, 154, Article 105004. <https://doi.org/10.1016/j.autcon.2023.105004>
- Weng, X., Yuan, C., Li, X., & Li, H. (2024). Research on the construction of a risk assessment indicator system for transportation infrastructure investment under public-private partnership model. *Buildings*, 14(6). <https://doi.org/10.3390/buildings14061679>
- Wisetjindawat, W., Kermanshah, A., Derrible, S., & Fujita, M. (2017). Stochastic modeling of road system performance during multihazard events: Flash floods and earthquakes [Article]. *Journal of Infrastructure Systems*, 23(4), Article 04017031. [https://doi.org/10.1061/\(ASCE\)IS.1943-555X.0000391](https://doi.org/10.1061/(ASCE)IS.1943-555X.0000391)
- Wu, C., Li, X., Guo, Y., Wang, J., Ren, Z., Wang, M., & Yang, Z. (2022). Natural language processing for smart construction: Current status and future directions. *Automation in Construction*, 134, Article 104059.
- Wu, C., Wu, P., Wang, J., Jiang, R., Chen, M., & Wang, X. (2021). Ontological knowledge base for concrete bridge rehabilitation project management. *Automation in Construction*, 121, Article 103428. <https://doi.org/10.1016/j.autcon.2020.103428>
- Xia, N., Wang, X., Wang, Y., Yang, Q., & Liu, X. (2017). Lifecycle cost risk analysis for infrastructure projects with modified Bayesian networks [Article]. *Journal of Engineering, Design and Technology*, 15(1), 79–103. <https://doi.org/10.1108/JEDT-05-2015-0033>
- Yazdani-Chamzini, A. (2014). Proposing a new methodology based on fuzzy logic for tunnelling risk assessment [Article]. *Journal of Civil Engineering and Management*, 20 (1), 82–94. <https://doi.org/10.3846/13923730.2013.843583>
- Zhong, B., Wu, H., Xiang, R., & Guo, J. (2022). Automatic information extraction from construction quality inspection regulations: A knowledge pattern-based ontological method. *Journal of Construction Engineering and Management*, 148(3). [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0002240](https://doi.org/10.1061/(ASCE)CO.1943-7862.0002240)
- Zhou, H., Tang, S., Huang, W., & Zhao, X. (2023). Generating risk response measures for subway construction by fusion of knowledge and deep learning. *Automation in Construction*, 152, Article 104951. <https://doi.org/10.1016/j.autcon.2023.104951>
- Zhu, F., Hu, H., Xu, F., & Tang, N. (2021). Predicting the impact of country-related risks on cost overrun for overseas infrastructure projects [Article]. *Journal of Construction Engineering and Management*, 147(2), Article 0012933. [https://doi.org/10.1061/\(ASCE\)CO.1943-7862.0001959](https://doi.org/10.1061/(ASCE)CO.1943-7862.0001959)
- Zou, P. X. W., Wang, S., & Fang, D. (2008). A life-cycle risk management framework for PPP infrastructure projects. *Journal of Financial Management of Property and Construction*, 13(2), 123–142. <https://doi.org/10.1108/13664380810898131>